06-Aug-2026

Scheduling of Projects – Introduction

CE716 : Project Management and Controls

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Department of Civil Engineering

Announcements

- Assignment #1 will be released today (Due date 13th Aug 2026)

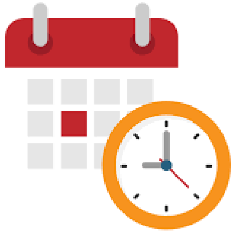
- Groups of 4

Why Schedule Projects?

Knowledge Required for Scheduling

Planning tools & software

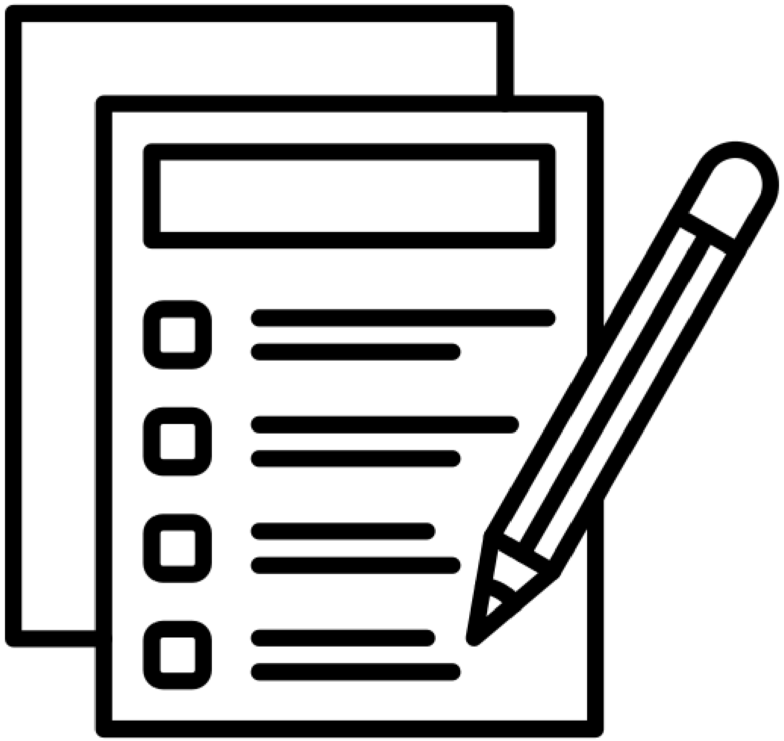
Knowledge of principles of scheduling and project management



Knowledge of specific technical field

Agenda

- Discussion on Assignment
- Work Breakdown Structure (WBS)
- Fundamentals of Scheduling
- Gantt Chart
- Practical applications (of Gantt chart)



Work-breakdown structure

- Breaking down of project into "manageable chunks"

- WBS represents "family-tree" of activities. (Useful for large projects)

- WBS provides common framework for planning, scheduling, cost estimating, budgeting, monitoring, reporting, directing, and controlling

- Typically,

  Project > Sub-project > Work-packages > Activities & Task

Work-breakdown structure

Definition as per PMBOK:

"a deliverable-oriented hierarchical decomposition of the work to be executed by the project team to accomplish the project objectives and create the required deliverables.

It organizes and defines the total scope of the project. Each descending level represents an increasingly detailed definition of the project work. The WBS is decomposed into work packages. The deliverable orientation of the hierarchy includes both internal and external deliverables."

How to breakdown projects?

- Based on the timeline
- Responsibility of person doing it
- Review time

How to breakdown projects?

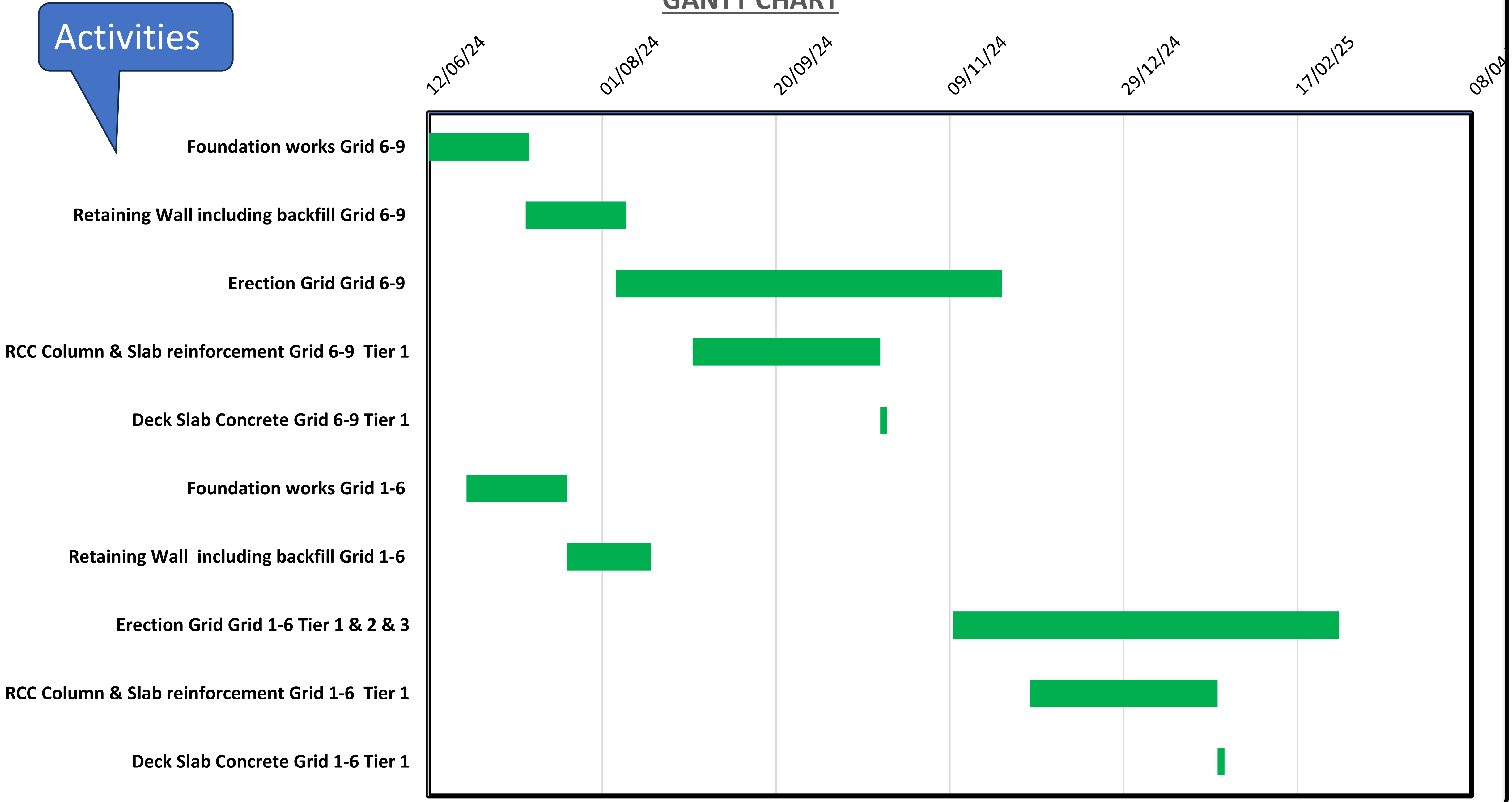
- No absolutely correct or incorrect way
- Always prefer a balanced approach
- The tasks/activities should be a reasonable number
- Activities should be easy to measure and control without being overly detailed.

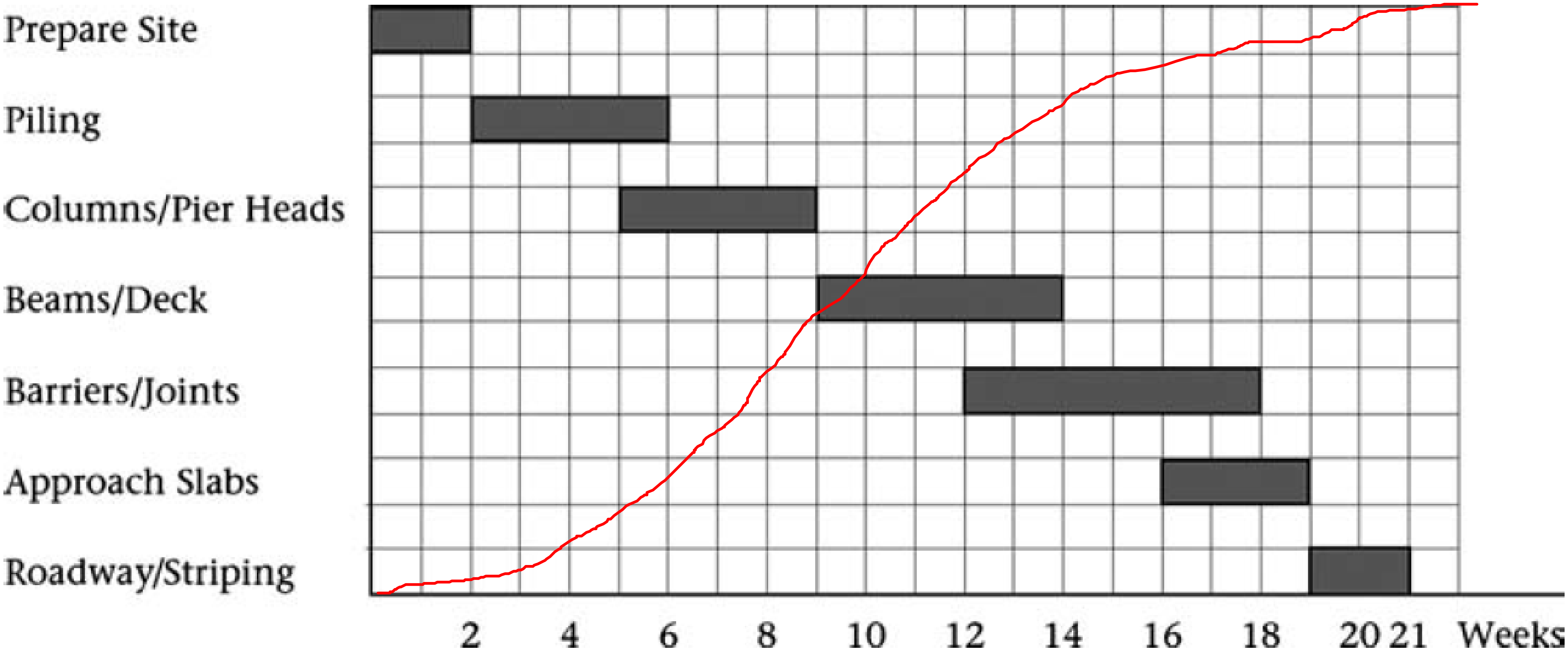
Gantt Chart or Bar Chart

Gantt Chart

- "A graphic representation of project activities, shown in a time-scaled bar"
- Developed by Henry L. Gantt in 1917
- Duration of each activity is determined – bars are drawn to show start and end of each activity
- "Links" between activities is not shown in Gantt Chart
- Bars may not indicate continuous work (its just the start & end)

GANTT CHART


Activities
12/06/24
01/08/24
20/09/24
09/11/24
29/12/24
17/02/25
08/04
Foundation works Grid 6-9
Retaining Wall including backfill Grid 6-9
Erection Grid Grid 6-9
RCC Column & Slab reinforcement Grid 6-9 Tier 1
Deck Slab Concrete Grid 6-9 Tier 1
Foundation works Grid 1-6
Retaining Wall including backfill Grid 1-6
Erection Grid Grid 1-6 Tier 1 & 2 & 3
RCC Column & Slab reinforcement Grid 1-6 Tier 1
Deck Slab Concrete Grid 1-6 Tier 1


Prepare Site
Piling
Columns/Pier Heads
Beams/Deck
Barriers/Joints
Approach Slabs
Roadway/Striping
2
4
6
8
10
12
14
16
18
20 21
Weeks

Potential Uses of Bar-chart (Gantt Chart)

- Simply show the start and end of each activity
- Display resources consumed along with activities
- Show cumulative budget along with activities
- With the as-built schedule
- Comparing as-planned schedule with the as-built schedule

Gantt Chart - Advantages

- Simple to draw and understand – Widely accepted across industries
- No “theory” or calculations needed – no trained/skilled personnel needed
- Activities are time-scaled. (Very easy to visualize duration of project)
- Preferred by non-technical people (managers/client)
- Can be used to load resource/money requirements to the chart
- Can be modified to show discontinuous activities
- Allows customization (rolling up of activities)

Gantt Chart – Disadvantages / Limitations

- Inability to depict interdependencies of activities (inability to incorporate logics)

- Difficult to answer – why did this activity start at this date?

(Logical constraint / resource constraint / subjective decision on PM)

- Absence of critical and non-critical activities

- Cost optimization or resource allocation is not possible

- Difficult to handle very large projects

Practical Applications (Discussion)

Is safety (time buffer) incorporated in the schedules?

How to make Tea (Black Tea)?

Activity	0	1	2	3	4	5	6	7	8	9	10
Find saucepan											
Boil water											
Find Tea leaves											
Add tea – let it simmer											
Filter and pour											

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If there is so much safety incorporated in the Schedule – Why do projects get delayed?

Incorporating Safety in Schedules

If there is so much safety incorporated in the Schedule – Why do projects get delayed?

- People who finish early won’t report it
- Even if they report – the person supposed to do the next step will believe there is sufficient time.
- The student’s syndrome (If there is more time- you tend to start late)
- Dependency in project – delays accumulate but advances get lost
- Impact of multi-tasking – its effect on lead time

Explore how to create Gantt Chart in Excel

THANK YOU